

MAKERERE UNIVERSITY BUSINESS SCHOOL

FINAL EXAMINATION FOR THE DEGREE OF

BACHELOR OF BUSINESS COMPUTING

OF MAKERERE UNIVERSITY ACADEMIC YEAR 2023/2024

PHOTOCOPIER

COURSE NAME: COMPUTING MATHEMATICS

YEAR OF STUDY: ONE

COURSE CODE: BUC 1114

SEMESTER: ONE

DATE: Saturday 9th December 2023

TIME: 9.00am – 12.00pm

INSTRUCTIONS:

1. This examination contains two sections, A and B.
2. Section A is compulsory and carries 40 marks.
3. Attempt only three questions in Section B

Section A (40 Marks). Attempt all questions

1. p is the proposition "Rebecca's PC has more than 16 GB free hard disk space" and q is the proposition "The processor in Rebecca's PC runs faster than 1 GHz. Write down the:
 - a. conjunction of p and q . (2 marks)
 - b. disjunction of p and q . (2 marks)
2. Given sets $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$. Using examples from sets A and B , describe the following set concepts
 - a. Subset (1 mark)
 - b. Power set (1 mark)
 - c. Cartesian product of A and B (1 mark)
 - d. Disjoint sets (1 mark)
3. Given the matrices $A = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 & 7 \\ 4 & 6 & 8 \end{bmatrix}$,
 - a. What is the order of matrices A and B ? (2 marks)
 - b. Work out
 - i. $A+B$ (1 mark)
 - ii. $3B$ (1 mark)
4. If $A = \{x \mid x \text{ is a positive integer between 3 and 8}\}$, work out the following.
 - a. Write set A in Roster notation (2 marks)
 - b. State the cardinality of set A (1 mark)
 - c. How many subsets does A have? (1 mark)

5. Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. Find
- $f \circ g$. (2 marks)
 - $g \circ f$. (2 marks)
6. Differentiate between a tautology and a contradiction (4 marks)
7. List down four properties of algorithms. (4 marks)
8. Assuming $S = \{1, 2, 3\}$ and R is the relation 'less than' on S . How many ordered pairs are in R ? (4 marks)
9. Define the negation of a proposition. What is the negation of "This is a boring course"? (4 marks)
10. Given vectors $u = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $v = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$ find
- $4u$ (1 mark)
 - $-3v$ (1 mark)
 - $w = 4u + (-3v)$ (1 mark)
 - $|w|$ (1 mark)

Section B (60 Marks). Attempt any THREE questions

QUESTION 11

- a) Evaluate the following:
- 479_{10} to Hexadecimal number system (04 marks)
 - -43_{10} to two's complement. (02 marks)
- b) Represent the decimal number 12.75 in IEEE 754 standard single precision. (08 marks)
- c) Explain three key applications of hexadecimal number system in Computers. (06 marks)

QUESTION 12

- a) Explain to your friend who has never stepped in a class of Computing mathematics the following terms as they are used in propositional logic.
- Syntax (2 marks)
 - Semantics (2 marks)
 - Truth Values (2 marks)
- b) Given the statement "There exists a prime number y that is even and divisible by 3".
- Form three predicates from the statement. (3 marks). $P(y)$: y is a prime number, $E(y)$: y is even, $D(y, 3)$: y is divisible by 3
 - Using logical symbols and the predicates above, express the statement in predicate logic. (3 marks)
- c) Your friend Lonitah thinks studying propositional logic is a waste of time for Business computing students. Her argument is that the focus should be on writing computer programs and learning how to use Business applications. As someone who is very knowledgeable in computing mathematics, briefly enlighten your friend on any four areas in computing where propositional logic is applied. (8 marks)

QUESTION 13

- a) The applicants for a certain vacant position at MUBS are given an aptitude test of 50 multiple-choice questions. Design an algorithm to output a list of the applicants

(identified by number), their scores, and a message stating whether they are to be short-listed for the position (those who score 36 or more) or reconsidered at a later date (those who score in the range from 0 to 35). The input consists of the answers provided by each applicant. (5 Marks)

b) Design an algorithm to get two numbers from the user (dividend and divisor), testing to make sure that the divisor number is not zero, and displaying their quotient using pseudocode (8 Marks)

c) Express the algorithm in (b) above using a flow chart (7 Marks)

QUESTION 14

$$S = 10 \quad W = 8$$

a. Given a group of students, 10 of them like soda (S), 8 like water (W), 6 like both soda and water, 2 like water only, 1 student likes neither soda nor water.

- Draw a clear Venn diagram to represent the information above. [2marks]
- Find the number of students who like soda only. [2marks]
- How many students are there in the group? [2marks]

b. Given the set $A = \{3, 4, 12, 24, 48, 72\}$ and the relation "is divisible".

- Find all the partial orders of set A. (4 marks)
- Draw a Hasse diagram showing all the partial orders listed above. (5 marks)

c. During a lecture, a student who missed the previous lecture sees the concept of sets and relations as abstract and have no practical application. The student asks the lecturer, if at all there are anywhere in the computing career that sets and relations are applied. As a student who was in that lecture, explain to your friend the various application of sets and relations are applied in computing. (5 marks)

QUESTION 15

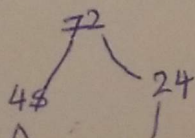
a. Given the system of linear equations $x - 2y = -1$ and $-x + 3y = 3$, answer the following questions

- Write the equations in matrix form $AX=B$, where A is the matrix coefficients, X the matrix of variables x and y, and B is the matrix of constants from the right sides of the equations (2 marks)
- Find the inverse of matrix A (4 marks)
- Find the product $A^{-1} \cdot B$ (4 marks)
- What is the solution to the equations? (2 marks)

b. Cite examples of applications where matrices are used in computing. (8 marks)

14(b) Partial orders
 $(3,3) (4,4) (12,12) (24,24) (48,48) (72,72)$
 $(4,12) (4,24) (4,48) (4,72)$
 $(3,12) (3,24) (3,48) (3,72)$
 $(12,24) (12,48) (12,72)$
 $(24,48) (24,72)$
 $(48,72)$

11 Hasse diagram



End of Question paper

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